HIGHER DIRECT IMAGES ON A RINGED SITE ARE THE SHEAF ASSOCIATED TO THE PRESHEAF OF COHOMOLOGY OF THE INVERSE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the module statement; it ends with the author's own closing phrase, which is retained verbatim.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology (ringed spaces) and in chapter etale-cohomology (schemes, small etale site). Both assert the same conclusion and both are special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted. The etale-cohomology twin additionally records a derived clause for K in D(X_etale); that clause is a corollary inside the special setting of the small etale site and is not counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Morphism of ringed sites and the inverse image functor u. Same chapter, the dictionary between morphisms of ringed sites and continuous functors u : D -> C. Supplies the functor u along which the presheaf V -> H^i(u(V), F) is formed.
- Higher direct images R^i f_* of a sheaf of modules. Same chapter, the definition of R^i f_* F as the i-th cohomology sheaf of the complex f_* I^bullet for an injective resolution F -> I^bullet. Supplies the object computed, and the injective resolution with which the counted proof starts.
- Sheafification and the abelian category structure on O_D-modules. Same chapter, the description of the cohomology sheaf of a complex of O_D-modules as the sheaf associated to the presheaf of kernel modulo image. Supplies the step in which the cohomology sheaf is written as a sheafified presheaf of sections.

Lemma 0.1. *Let $f : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{D}, \mathcal{O}_\mathcal{D})$ be a morphism of ringed sites corresponding to the continuous functor $u : \mathcal{D} \to \mathcal{C}$. For any $\mathcal{F} \in \mathrm{Ob}(Mod(\mathcal{O}_\mathcal{C}))$ the sheaf $R^i f_* \mathcal{F}$ is the sheaf associated to the presheaf*

$$V \longmapsto H^i(u(V), \mathcal{F})$$

Proof. Let $\mathcal{F} \to \mathcal{I}^\bullet$ be an injective resolution. Then $R^i f_* \mathcal{F}$ is by definition the ith cohomology sheaf of the complex

$$f_* \mathcal{I}^0 \to f_* \mathcal{I}^1 \to f_* \mathcal{I}^2 \to \ldots$$

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By definition of the abelian category structure on $\mathcal{O}_\mathcal{D}$-modules this cohomology sheaf is the sheaf associated to the presheaf

$$V \longmapsto \frac{\mathrm{Ker}(f_*\mathcal{I}^i(V) \to f_*\mathcal{I}^{i+1}(V))}{\mathrm{Im}(f_*\mathcal{I}^{i-1}(V) \to f_*\mathcal{I}^i(V))}$$

and this is obviously equal to

$$\frac{\mathrm{Ker}(\mathcal{I}^i(u(V)) \to \mathcal{I}^{i+1}(u(V)))}{\mathrm{Im}(\mathcal{I}^{i-1}(u(V)) \to \mathcal{I}^i(u(V)))}$$

which is equal to $H^i(u(V), \mathcal{F})$ and we win. $\square$